

LAB SESSION 1

Programming Exercise:

Question 1: Calculate Area of a Rectangle: Write a C++ program to calculate the area of a rectangle given its length and width.

```
C: > C++ PROGRAMS > C++ tut1.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int length, width;
7      cout << "Enter length: ";
8      cin >> length;
9      cout << "Enter width: ";
10     cin >> width;
11
12     int area = length * width;
13     cout << "Area of the rectangle: " << area << endl;
14
15     return 0;
16 }
17
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter length: 3
Enter width: 5
Area of the rectangle: 15
PS C:\C++ PROGRAMS>
```

Question 2: Check Even or Odd: Write a C++ program to check if a given number is even or odd.

```
C: > C++ PROGRAMS > C++ tut1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int number;
7      cout << "Enter a number: ";
8      cin >> number;
9
10     if (number % 2 == 0)
11     {
12         cout << number << " is even" << endl;
13     }
14     else
15     {
16         cout << number << " is odd" << endl;
17     }
18
19     return 0;
20 }
21
```

```
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 7
7 is odd
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 2
2 is even
PS C:\C++ PROGRAMS>
```

Question 3: Generate Fibonacci Series: Write a C++ program to generate the Fibonacci series up to a given number of terms.

```
C:\C++ PROGRAMS > C++ tut1.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int n;
7      cout << "Enter the number of terms: ";
8      cin >> n;
9
10     int a = 0, b = 1, next;
11
12     cout << "Fibonacci Series: ";
13
14     for (int i = 0; i < n; i++)
15     {
16         cout << a << " ";
17         next = a + b;
18         a = b;
19         b = next;
20     }
21
22     return 0;
23 }
24
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter the number of terms: 10
Fibonacci Series: 0 1 1 2 3 5 8 13 21 34
PS C:\C++ PROGRAMS>
```

Question 4: Find Maximum Number: Write a C++ program to find the maximum number among three given numbers.

```
C:\C++ PROGRAMS > C++ tut1.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int num1, num2, num3;
7      cout << "Enter three numbers: ";
8      cin >> num1 >> num2 >> num3;
9
10     int maxNum;
11
12     if (num1 >= num2 && num1 >= num3)
13     {
14         maxNum = num1;
15     }
16     else if (num2 >= num1 && num2 >= num3)
17     {
18         maxNum = num2;
19     }
20     else
21     {
22         maxNum = num3;
23     }
24
25     cout << "Maximum number: " << maxNum << endl;
26
27     return 0;
28 }
29
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter three numbers: 10
20
15
Maximum number: 20
PS C:\C++ PROGRAMS>
```

Question 5: Reverse a String: Write a C++ program to reverse a given string.

```
C:\C++ PROGRAMS > C++ tut1.cpp > ...
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main()
6  {
7      string str;
8      cout << "Enter a string: ";
9      cin >> str;
10
11     string reversed = "";
12
13     for (int i = str.length() - 1; i >= 0; i--)
14     {
15         reversed += str[i];
16     }
17
18     cout << "Reversed string: " << reversed << endl;
19
20     return 0;
21 }
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a string: hello
Reversed string: olleh
PS C:\C++ PROGRAMS>
```

Question 6: Check Prime Number: Write a C++ program to check if a given number is prime or not.

```
C:\C++ PROGRAMS > C++ tut1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int number;
7      bool isPrime = true;
8
9      cout << "Enter a number: ";
10     cin >> number;
11     if (number <= 1)
12     {
13         isPrime = false;
14     }
15     else
16     {
17         for (int i = 2; i <= number / 2; i++)
18         {
19             if (number % i == 0)
20             {
21                 isPrime = false;
22                 break;
23             }
24         }
25         if (isPrime)
26             cout << number << " is prime" << endl;
27         else
28             cout << number << " is not prime" << endl;
29         return 0;
30     }
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 7
7 is prime
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 4
4 is not prime
PS C:\C++ PROGRAMS>
```

Question 7: Calculate Factorial: Write a C++ program to calculate the factorial of a given number.

```
C:\C++ PROGRAMS > g++ tut1.cpp & main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int number;
7      unsigned long long factorial = 1;
8
9      cout << "Enter a number: ";
10     cin >> number;
11
12     if (number < 0)
13     {
14         cout << "Factorial is not defined for negative numbers." << endl;
15     }
16     else
17     {
18         for (int i = 1; i <= number; i++)
19         {
20             factorial *= i;
21         }
22         cout << "Factorial of " << number << ": " << factorial << endl;
23     }
24
25     return 0;
26 }
```

```
PS C:\Users\hnp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 5
Factorial of 5: 120
PS C:\C++ PROGRAMS>
```

Question 8: Calculate Simple Interest: Write a C++ program to calculate the simple interest given the principal amount, rate of interest, and time period.

```
C:\C++ PROGRAMS > g++ tut1.cpp & main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      float principal, rate, time, simpleInterest;
7
8      cout << "Enter principal amount: ";
9      cin >> principal;
10
11     cout << "Enter rate of interest: ";
12     cin >> rate;
13
14     cout << "Enter time (in years): ";
15     cin >> time;
16
17     simpleInterest = (principal * rate * time) / 100;
18
19     cout << "Simple interest: " << simpleInterest << endl;
20
21     return 0;
22 }
23
```

```
PS C:\Users\hnp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter principal amount: 100
Enter rate of interest: 5
Enter time (in years): 3
Simple interest: 15
PS C:\C++ PROGRAMS>
```

Question 9: Check Leap Year: Write a C++ program to check if a given year is a leap year or not.

```
C:\C++ PROGRAMS > C++ tut1.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int year;
7      cout << "Enter a year: ";
8      cin >> year;
9
10     if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
11     {
12         cout << year << " is a leap year" << endl;
13     }
14     else
15     {
16         cout << year << " is not a leap year" << endl;
17     }
18
19     return 0;
20 }
21
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a year: 2024
2024 is a leap year
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a year: 1999
1999 is not a leap year
PS C:\C++ PROGRAMS>
```

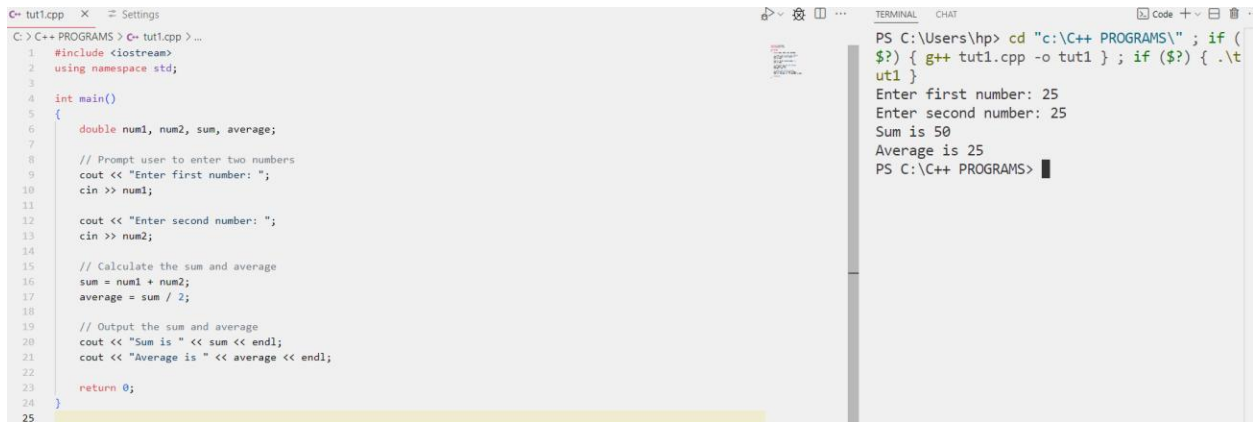
Question 10: Check Armstrong Number: Write a C++ program to check if a given number is an Armstrong number or not.

```
C:\C++ PROGRAMS > C++ tut1.cpp > main()
1  #include <iostream>
2  #include <cmath>
3  using namespace std;
4  int main()
5  {
6      int number, sum = 0, temp, remainder, digits = 0;
7
8      cout << "Enter a number: ";
9      cin >> number;
10
11      temp = number;
12
13      // Calculate the number of digits in the number
14      while (temp != 0)
15      {
16          temp /= 10;
17          digits++;
18      }
19      temp = number;
20
21      // Calculate the sum of the power of each digit
22      while (temp != 0)
23      {
24          remainder = temp % 10;
25          sum += pow(remainder, digits);
26          temp /= 10;
27      }
28
29      if (sum == number)
30      {
31          cout << number << " is an Armstrong number" << endl;
32      }
33      else
34      {
35          cout << number << " is not an Armstrong number" << endl;
36      }
37      return 0;
38 }

```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 153
153 is an Armstrong number
PS C:\C++ PROGRAMS> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a number: 148
148 is not an Armstrong number
PS C:\C++ PROGRAMS>
```

Question 11: Sum and Average Calculation: Write a C++ program that prompts the user to enter two numbers, calculates their sum, and then computes their average.



```
C++ tut1.cpp X Settings
C:\> C++ PROGRAMS > C- tut1.cpp > ...
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     double num1, num2, sum, average;
7
8     // Prompt user to enter two numbers
9     cout << "Enter first number: ";
10    cin >> num1;
11
12    cout << "Enter second number: ";
13    cin >> num2;
14
15    // Calculate the sum and average
16    sum = num1 + num2;
17    average = sum / 2;
18
19    // Output the sum and average
20    cout << "Sum is " << sum << endl;
21    cout << "Average is " << average << endl;
22
23    return 0;
24 }
25
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter first number: 25
Enter second number: 25
Sum is 50
Average is 25
PS C:\C++ PROGRAMS>
```

Question 12: Time Conversion: Write a C++ program that prompts the user to enter time in seconds and then converts it into hours, minutes, and seconds.



```
C:\C++ PROGRAMS > C- tut1.cpp > ...
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int totalSeconds, hours, minutes, seconds;
7
8     // Prompt user to enter time in seconds
9     cout << "Enter the time in seconds: ";
10    cin >> totalSeconds;
11
12    // Calculate hours, minutes, and seconds
13    hours = totalSeconds / 3600;
14    totalSeconds %= 3600;
15    minutes = totalSeconds / 60;
16    seconds = totalSeconds % 60;
17
18    // Output the result
19    cout << "Hours in time is: " << hours << endl;
20    cout << "Minutes in time: " << minutes << endl;
21    cout << "Seconds in time is: " << seconds << endl;
22
23    return 0;
24 }
25
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\" ; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter the time in seconds: 3713
Hours in time is: 1
Minutes in time: 1
Seconds in time is: 53
PS C:\C++ PROGRAMS>
```

Question 13: Amount in Rupees Breakdown: Write a C++ program that prompts the user to enter an amount in rupees and then breaks it down into denominations of 1000s, 500s, 100s, 50s, 10s, 5s, 2s, and 1s.

```

1 #include <iostream>
2 using namespace std;
3 int main()
4 {
5     int amount;
6
7     cout << "Enter amount in rupees: ";
8     cin >> amount;
9
10    int thousands = amount / 1000;
11    amount %= 1000;
12
13    int fiveHundreds = amount / 500;
14    amount %= 500;
15
16    int hundreds = amount / 100;
17    amount %= 100;
18
19    int fifties = amount / 50;
20    amount %= 50;
21
22    int tens = amount / 10;
23    amount %= 10;
24
25    int fives = amount / 5;
26    amount %= 5;
27
28    int twos = amount / 2;
29    amount %= 2;
30
31    int ones = amount;
32    cout << "1000's in the given amount is: " << thousands << endl;
33    cout << "500's in the given amount is: " << fiveHundreds << endl;
34    cout << "100's in the given amount is: " << hundreds << endl;
35    cout << "50's in the given amount is: " << fifties << endl;
36    cout << "10's in the given amount is: " << tens << endl;
37    cout << "5's in the given amount is: " << fives << endl;
38    cout << "2's in the given amount is: " << twos << endl;
39    cout << "1's in the given amount is: " << ones << endl;
40    return 0;
41 }

```

```

$?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\t
ut1 }
Enter amount in rupees: 5718
1000's in the given amount is: 5
500's in the given amount is: 1
100's in the given amount is: 2
50's in the given amount is: 0
10's in the given amount is: 1
5's in the given amount is: 1
2's in the given amount is: 1
1's in the given amount is: 1
PS C:\C++ PROGRAMS>

```

Question 14: Fahrenheit to Celsius Conversion: Write a C++ program that prompts the user to enter a temperature in Fahrenheit and then converts it into Celsius using the formula: Celsius = (Fahrenheit - 32) * 5 / 9.

```

1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     float fahrenheit, celsius;
7
8     // Prompt user to enter temperature in Fahrenheit
9     cout << "Enter the Fahrenheit temperature: ";
10    cin >> fahrenheit;
11
12    // Convert Fahrenheit to Celsius
13    celsius = (fahrenheit - 32) * 5 / 9;
14
15    // Output the Celsius temperature
16    cout << "Celsius Temperature: " << celsius << endl;
17
18    return 0;
19 }
20

```

```

$?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\t
ut1 }
Enter the Fahrenheit temperature: 98.6
Celsius Temperature: 37
PS C:\C++ PROGRAMS>

```

Question 15: Two-Digit Integer Reversal: Write a C++ program that inputs a two-digit integer value and outputs its reverse order.

```
C:\C++ PROGRAMS > g++ tut1.cpp > ...
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int number, tens, ones;
7
8      // Prompt user to enter a two-digit integer
9      cout << "Enter a 2 digit integer value: ";
10     cin >> number;
11
12     // Extract tens and ones places
13     tens = number / 10; // Extract the tens place
14     ones = number % 10; // Extract the ones place
15
16     // Reverse the number
17     int reversed = (ones * 10) + tens;
18
19     // Output the reversed number
20     cout << "Reverse of this value is: " << reversed << endl;
21
22     return 0;
23 }
```

```
PS C:\Users\hp> cd "c:\C++ PROGRAMS\"; if ($?) { g++ tut1.cpp -o tut1 } ; if ($?) { .\tut1 }
Enter a 2 digit integer value: 54
Reverse of this value is: 45
PS C:\C++ PROGRAMS>
```